

Multipolar polarizabilities with finite surface impedance

The focus of these notes is to derive **equation 9** in the paper: [Localized plasmons in graphene-coated nanospheres](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414) (<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.91.125414>), which was also used, without derivation, in the notes by us (https://hackmd.io/@aligho/ByRI-iJ_ge) (https://hackmd.io/@aligho/ByRI-iJ_ge). **equation 9** gives the multipolar polarizability in the quasistatic limit. The paper [Multipolar response of small metallic spheres: Nonlocal theory](https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722) (<https://journals.aps.org/prb/pdf/10.1103/PhysRevB.35.3722>) is also useful. In the following, ϵ_m is the dielectric function of the sphere, a its radius, and ϵ_d the dielectric function of the environment (in the first part of the derivation we assume $\epsilon_d = 1$). SI units are used throughout. In addition, for the coefficients of the fields, we use subscripts, $<$, $>$, to denote quantities pertaining to fields inside and outside the scatterer, respectively.

We take the electric field to be the gradient of a scalar potential, $\mathbf{E}(\mathbf{r}) = -\nabla\phi(\mathbf{r})$. Since we require $\nabla^2\phi(\mathbf{r}) = 0$ everywhere, we know that we may take

$$\phi(r, \theta, \phi) = \sum_l \left[\frac{a_l}{r^{l+1}} + b_l r^l \right] P_l(\cos(\theta))$$

We have two boundary conditions on the scalar potential and we also know that it may not diverge at the origin, which necessarily implies:

$$b_{l,<} a^l = \frac{a_{l,>}}{a^{l+1}} + b_{l,>} a^l$$

$$\epsilon_m l b_{l,<} a^{l-1} = -\frac{(l+1)a_{l,>}}{a^{l+2}} + l b_{l,>} a^{l-1},$$

Where the first boundary condition corresponds to the continuity of the tangential component of the electric field (equivalently, continuity of the potential), and the second boundary condition corresponds to the continuity of the normal component of

the displacement field. Without loss of generality, we set $b_{l,>}$ to unity (corresponding to an overall rescaling of all fields). Our goal is to find $a_{l,>}$:

$$\begin{aligned} \varepsilon_m l \left[\frac{a_{l,>}}{a^{2l+1}} + 1 \right] &= -\frac{(l+1)a_{l,>}}{a^{2l+1}} + l \rightarrow a_{l,>} \left[\varepsilon_m l + (l+1) \right] = a^{2l+1} l [1 - \varepsilon_m] \\ &\rightarrow a_{l,>} = a^{2l+1} \frac{l(1 - \varepsilon_m)}{(l+1) + \varepsilon_m l} \end{aligned}$$

The polarizability is the ratio between the induced dipole moment and the applied field. The moment is given by $4\pi\varepsilon_0 a_{l,>}$ (which we prove in the appendix) and the applied field is given by $-b_l$. Therefore, we get that the multipolar polarizability is given by:

$$4\pi\varepsilon_0 a^{2l+1} \frac{l(\varepsilon_m - 1)}{(l+1) + \varepsilon_m l}$$

Now we consider the effect of non-vanishing surface impedance. In this case, the displacement field is not continuous across the boundary. This is because the finite surface conductivity $\sigma(\omega)$ yields a non-zero charge density given implicitly by the continuity equation:

$$-i\omega\rho(\omega) + \nabla \cdot \mathbf{J} = 0 \rightarrow \rho(\omega) = -\frac{i}{\omega} \nabla \cdot \mathbf{J},$$

where the current density is given by:

$$\mathbf{J}(r, \cos(\theta)) = \sigma(\omega) E_\theta \delta(r - a) e_\theta = -\sigma(\omega) (\nabla \phi) \cdot e_\theta \delta(r - a) e_\theta$$

Note that the divergence in spherical coordinates may be written as:

$$\frac{ib_{l,<} a^l \sigma(\omega)}{\omega a^2 \sin(\theta)} \partial_\theta (\sin(\theta) \partial_\theta P_l(\cos(\theta))) \delta(r - a)$$

Now, note that Legendre polynomials satisfy the equation:

$$\left[(1 - x^2) P'_l(x) \right]' = -l(l+1) P_l(x)$$

Since we may write: $\partial_x = -(\sin(\theta))^{-1}\partial_\theta$, we may rewrite the above as:

$$\frac{1}{\sin(\theta)}\partial_\theta \left[\sin(\theta)\partial_\theta P_l(\cos(\theta)) \right] = -l(l+1)P_l(\cos(\theta))$$

This gives us, for the induced charge density (per unit area):

$$-\frac{i\sigma(\omega)l(l+1)a^l b_{l,<}}{\omega a^2} P_l(\cos(\theta))$$

Therefore, our generalized boundary conditions are given by:

$$\begin{aligned} b_{l,<}a^l &= \frac{a_{l,>}}{a^{l+1}} + a^l \\ \varepsilon_d \left[\frac{(l+1)a_{l,>}}{a^{l+2}} - la^{l-1} \right] + \varepsilon_m l b_{l,<}a^{l-1} &= -\frac{i\sigma(\omega)l(l+1)}{\varepsilon_0 \omega a^2} a^l b_{l,<} \rightarrow \\ -\frac{i\sigma(\omega)l(l+1)}{\varepsilon_0 \omega a^2} \left[\frac{a_{l,>}}{a^{l+1}} + a^l \right] &= \varepsilon_d \left[\frac{(l+1)a_{l,>}}{a^{l+2}} - la^{l-1} \right] + \varepsilon_m l \left[\frac{a_{l,>}}{a^{l+2}} + a^{l-1} \right] \end{aligned}$$

We first multiply both sides by a^{l+2} , which gives us:

$$-\frac{i\sigma(\omega)l(l+1)}{\varepsilon_0 \omega a} \left[a_{l,>} + a^{2l+1} \right] = \varepsilon_d \left[(l+1)a_{l,>} - la^{2l+1} \right] + \varepsilon_m l \left[a_{l,>} + a^{2l+1} \right]$$

Collecting terms gives us:

$$a_{l,>} \left[-l\varepsilon_m - (l+1)\varepsilon_d - g(\omega)l(l+1) \right] = a^{2l+1} \left[l\varepsilon_m - l\varepsilon_d + g(\omega)l(l+1) \right]$$

Thus, we get the non-retarded polarizability:

$$\alpha = \frac{4\pi\varepsilon_0 l \left[\varepsilon_m - \varepsilon_d + (l+1)g(\omega) \right]}{l\varepsilon_m + (l+1)\varepsilon_d + g(\omega)l(l+1)}$$

Consistency with retarded result

We use expansions of the spherical Bessel function from **equations 5.1, 5.2** in **Bohren and Huffmann**. Each term in the denominator of the TM scattering coefficients involves a product of $j_l(x_1)$ or $h_l^1(x_2)$ (or their derivatives). Since we are concerned with the small x_1, x_2 limit, we need only consider the $y_l(x_2)$ term in the Hankel function. Note that the notation used below is written consistently with the Christensen paper.

The denominator of the TM scattering coefficients reads (considering only the most divergent components):

$$x_2^2 y_l(x_2) [x_1 j_l(x_1)]' - x_1^2 j_l(x_1) [x_2 y_l(x_2)]' - g(\omega) x_0^2 [x_1 j_l(x_1)]' [x_2 y_l(x_2)]'$$

Our approximations for the various spherical bessel functions are given by:

$$j_l(x_1) = \frac{x_1^l}{(2l+1)!!}, [x_1 j_l(x_1)]' = \frac{(l+1)x_1^l}{(2l+1)!!}$$

$$y_l(x_2) = -\frac{(2l-1)!!}{x_2^{l+1}}, [x_2 y_l(x_2)]' \approx \frac{l(2l-1)!!}{x_2^{l+1}}$$

Therefore, we may write the denominator of the TM scattering coefficients as (up to a constant of proportionality):

$$(l+1)x_1^l x_2^{-l+1} + l x_1^{l+2} x_2^{-l-1} + g(\omega) x_0^2 l(l+1) x_1^l x_2^{-l-1} =$$

$$x_1^l x_2^{-l-1} \left[(l+1)x_2^2 + l x_1^2 + g(\omega) x_0^2 l(l+1) \right]$$

We now write everything in terms of x_0 , i.e. $x_1 = \sqrt{\varepsilon_m} x_0$ and $x_2 = \sqrt{\varepsilon_d} x_0$, which gives us (again, up to a proportionality factor):

$$(l+1)\varepsilon_d + l\varepsilon_m + g(\omega)l(l+1)$$

Therefore, since the plasmon dispersion is given by the poles of the scattering coefficients (or the poles of the polarizability), we see that in the small ka limit, the quasistatic result is consistent with the fully retarded result. Note that without surface impedance, and with a vacuum/air environment ($\varepsilon_d = 1$), in the true $ka \rightarrow 0$ limit, we have CPA at $\varepsilon_m = -(l+1)/l$. Therefore, for $l = 1$, $\varepsilon_m = -2$, for $l = 2$, $\varepsilon_m = -3/2$,

etc. This is consistent with the plots in the figures shown in [Perfect coupling of light to surface plasmons by coherent absorption](#)

(<https://journals.aps.org/prl/pdf/10.1103/PhysRevLett.108.186805>) (Fig 2(a,b)).

TE plasmon modes

The TE plasmon modes cannot be found from the nonretarded polarizability, but we may find the condition for their existence in the small ka limit. In this case, the denominator of the TE scattering coefficients (the most divergent part) is given by:

$$j_l(x_1)[x_2 y_l(x_2)]' - y_l(x_2)[x_1 j_l(x_1)]' + g(\omega)x_0^2 j_l(x_1)y_l(x_2)$$

$$lx_1^l x_2^{-l-1} + (l+1)x_2^{-l-1}x_1^l - g(\omega)x_0^2 x_1^l x_2^{-l-1}$$

Therefore, for a TE plasmon to exist, we require: $2l+1 - g(\omega)x_0^2 = 0$. Equivalently,

$$\frac{\sigma(\omega)}{\omega \epsilon_0 a} = \frac{(2l+1)c^2}{\omega^2 a^2} \rightarrow \sigma(\omega) = -i \frac{(2l+1)}{\mu_0 \omega a}$$

Note that ωa has units of $(\epsilon_0 \mu_0)^{-1/2}$, which means that $\sigma(\omega)$ will have units of $(\epsilon_0/\mu_0)^{1/2}$ (inverse of free space impedance, as required).

Numerical evaluation of the radiative corrections

As we showed in https://hackmd.io/@aligho/ByRI-iJ_ge (https://hackmd.io/@aligho/ByRI-iJ_ge), if we take only the intraband conductivity into account, we may write:

$$g(\omega) = -800\alpha\epsilon_F/[(\hbar\omega)^2 a],$$

where a is given in nanometers, $\hbar\omega$ and ϵ_F are given in electronvolts. The denominator of the TM reflection coefficients is given, for each angular momentum, l ,

$$x_2^2 h_l^1(x_2)[x_1 j_l(x_1)]' - x_1^2 j_l(x_1)[x_2 h_l(x_2)]' - g(\omega)x_0^2[x_1 j_l(x_1)]'[x_2 h_l^1(x_2)]',$$

where $x_0 = x_2 = a\omega/c$ and $x_1 = \epsilon_m a\omega/c$

We write $a\omega/c$ as $a\hbar\omega/(\hbar c)$. Since we give a in nanometers, we write $\hbar c \approx 200\text{eV} \times \text{nm}$, so that we may write $x_0 = x_2 = a\hbar\omega/200$, where a is given in nanometers and $\hbar\omega$ is given in eV.

Appendix 1: Multipolar polarizability in terms of the field outside the scatterer

We define the l 'th multipole of the charge distribution, $\rho(r, \theta)$ as follows (see https://en.wikipedia.org/wiki/Spherical_multipole_moments

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$$\sqrt{\frac{4\pi}{2l+1}} \int Y_{lm}^*(\theta, \phi) r^l \rho(r, \theta, \phi) d^3\mathbf{r} = - \int P_l(\cos(\theta)) r^{l+2} \rho(r, \theta) dr (2\pi) d(\cos(\theta)),$$

where in the second equality we used the relation between the spherical harmonics and the Legendre polynomials, valid for azimuthally symmetric charge distributions. We know that the induced charge density only exists on the boundary (i.e.

$\rho(r, \theta) = \rho(\theta)\delta(r - a) \equiv \rho P_l(\cos(\theta))\delta(r - a)$). With this in mind (and also noting that the Legendre polynomials are normalized such that $\int_{-1}^1 P_l^2(x) dx = 2/(2l+1)$), we have that the multipole moment is given by:

$$- \int P_l(\cos(\theta)) r^{l+2} \rho(r, \theta) dr (2\pi) d(\cos(\theta)) = \frac{4\pi a^{l+2}}{2l+1} \rho_l$$

At this point, all we need to do is calculate ρ_l in terms of the field quantities. To do so, we note that ρ_l is directly related to the radial change of the electric field over the surface of the sphere (as in the main text, we set $b_{l,>} = 1$):

$$\frac{\rho_l}{\epsilon_0} = \frac{(l+1)a_{l,>}}{a^{l+2}} + la^{l-1}(b_{l,<} - 1)$$

At this point, we are almost done. However, we wish to express ρ_l entirely in terms of $a_{l,>}$. To do so, we use the continuity of the tangential part of the electric field to write

$$b_{l,<}a^{l-1} = \frac{a_{l,>}}{a^{l+2}} + a^{l-1} \rightarrow l(b_{l,<} - 1)a^{l-1} = \frac{la_{l,>}}{a^{l+2}}$$

Therefore, we have, immediately,

$$\rho_l/\varepsilon_0 = \frac{(2l+1)a_{l,>}}{a^{l+2}},$$

giving us a multipole moment $4\pi\varepsilon_0 a_{l,>}$, as stated in the main text.